

Check this GCC and Consumption layer segregation of the sizing sheet

Worked for 36s >

Yes. I checked the **GCC vs Consumption Layer segregation** across the DC, DR, UAT and Pre-Prod rows in the sizing sheet.

1. What the sheet is currently doing

The sheet is using this logic:

- **GCC** → core platform / shared infrastructure
- **Consumption Layer** → services/workloads that consume GCC compute/storage

This is a **reasonable segregation**, but I see **one important item that should be reviewed: Keycloak**.

Component / Service	Current classification	My recommendation	Reason
K8 Master	GCC	✓ GCC	Kubernetes control-plane/shared platform component
K8 Worker	GCC	✓ GCC*	Correct if workers are dedicated to the shared Kubernetes/platform layer
Monitoring	GCC	✓ GCC	Shared infrastructure/platform monitoring
Apache NiFi	Consumption	✓ Consumption	Data ingestion workload

Component / Service	Current classification	My recommendation	Reason
Kafka	Consumption	✓ Consumption	Data/event streaming workload
Airflow	Consumption	✓ Consumption	Data workflow/orchestration workload
Spark Master	Consumption	✓ Consumption	Data processing workload
Spark Worker	Consumption	✓ Consumption	Data processing workload
Trino Coordinator	Consumption	✓ Consumption	Query/analytics workload
OpenSearch	Consumption	✓ Consumption	Search/analytics workload
H200 / AI Compute	Consumption	✓ Consumption	AI/ML workload
PostgreSQL	Consumption	✓ Consumption	Application/data workload
MongoDB	Consumption	✓ Consumption	Application/data workload
Milvus	Consumption	✓ Consumption	Vector database / AI workload
Neo4j	Consumption	✓ Consumption	Graph database / analytics workload
MinIO / Object Storage	Consumption	⚠ Depends	Depends whether it is shared platform storage or data-lake storage
Keycloak	Consumption	⚠ Review → preferably GCC	Identity & access management is generally a shared platform service

2. The main correction I would make

● Keycloak

In your sheet, Keycloak is classified as:

Consumption Layer – Service consuming GCC compute/storage resources

I would **prefer putting Keycloak in GCC** if the same Keycloak instance is being used for authentication/authorization across the entire platform.

Why?

Think of GCC as the **common infrastructure layer**:

GCC

- Kubernetes control plane
- Shared Kubernetes workers
- Monitoring
- Identity & Access Management
- Shared platform services

Then:

Consumption Layer

- NiFi
- Kafka
- Airflow
- Spark
- Trino
- OpenSearch
- Databases
- Vector DB
- AI/ML compute
- Data workloads

Keycloak is not really a business/data workload. It is a **common security service used by those workloads**.

So I would change:

Keycloak → GCC

provided the architecture has **one shared Keycloak service for the platform**.

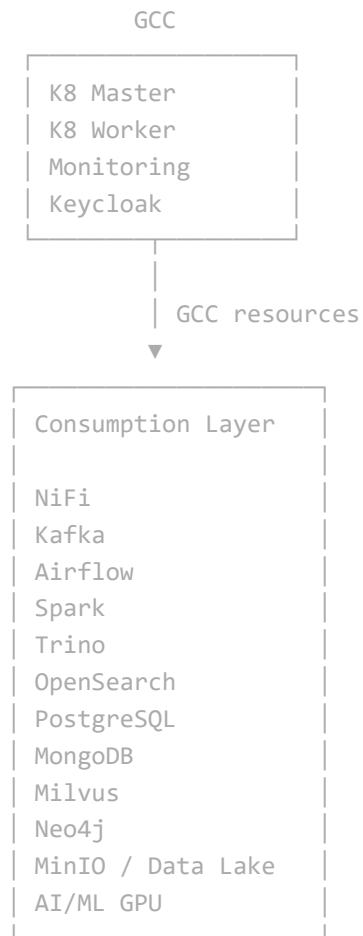
3. MinIO needs clarification

Your sheet has:

S1 → Object Storage → MinIO → Consumption Layer

This can be correct if MinIO is the actual Data Lake/Object Storage used by the consumption workloads.

For example:



In this scenario, **MinIO belongs in Consumption** because it is providing the actual data-lake/object-storage capacity consumed by the data platform.

However, if MinIO is intended to be **shared infrastructure storage for GCC itself**, then it should be GCC.

So this needs confirmation from the OEM/vendor.

4. Important point about K8 Worker

The sheet currently has:

K8 Master → GCC

K8 Worker → GCC

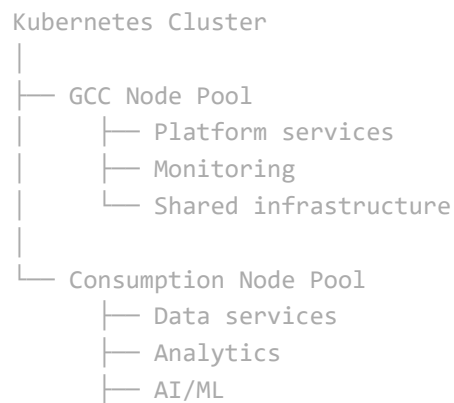
This is acceptable **only if these worker nodes are dedicated to GCC/shared platform services**.

If the same Kubernetes workers will actually run:

- NiFi
- Kafka
- Airflow
- Spark
- Trino
- Keycloak
- etc.

then simply categorizing the entire worker capacity as GCC becomes questionable.

A better architecture would be:



└─ Application workloads

Or separate clusters if the RFP/security requirements demand stronger isolation.

This is one of the questions I would definitely ask the OEM.

5. My recommended final segregation

GCC

Component

K8 Master



K8 Worker*



Monitoring



Keycloak



Other shared platform/security services



Consumption Layer

Component

Consumption

Apache NiFi



Kafka



Airflow



Spark Master



Spark Worker



Trino Coordinator



Component	Consumption
OpenSearch	✓
H200 / AI GPU	✓
PostgreSQL	✓
MongoDB	✓
Milvus	✓
Neo4j	✓
MinIO / Data Lake Object Storage*	✓

6. Overall assessment of your sheet

Current segregation: ~85–90% conceptually correct.

I would **not reject the sizing sheet**, but I would send it back to the OEM with these clarifications:

1. Why is Keycloak classified under Consumption Layer instead of GCC/shared platform?
2. Is MinIO the Data Lake/object storage or shared GCC infrastructure storage?
3. Are K8 Worker nodes dedicated exclusively to GCC/platform services?
4. If consumption workloads run on Kubernetes, where is their worker/node-pool capacity accounted for?
5. Are GCC and Consumption Layer physically separated, logically separated, or only separated through VM/resource allocation?
6. Are the same GCC services shared across DC, DR, UAT and Pre-Prod, or does each environment have dedicated instances?

One particularly important observation

Your sheet repeats the GCC components in every environment:

- PROD-DC
- PROD-DR
- UAT
- Pre-Prod

That is fine if the RFP requires environment-wise isolation. But if GCC is intended as a **central/shared platform**, you should verify why separate K8 Master/Worker and Monitoring capacity is being sized for each environment.

So before accepting the sizing, I would specifically ask the OEM to provide a **GCC vs Consumption architecture mapping showing which VM/node belongs to which layer and why**. This will make the sizing much easier to validate.



Yes. Based on the **GCC vs Consumption Layer segregation in this sizing sheet**, I would raise the following questions to the vendor. The first 6 are the most important.

Questions to Vendor – GCC & Consumption Layer

#	Question to Vendor	Why we should ask
1	Please provide the detailed mapping of each component/service to GCC or Consumption Layer, along with the architectural justification for the segregation.	To ensure the sizing classification is not arbitrary.
2	Keycloak is currently classified under the Consumption Layer. Please clarify why it is not considered a shared GCC/platform service, given that it provides common IAM/authentication services to multiple platform components.	Keycloak normally acts as a shared platform/security service.

#	Question to Vendor	Why we should ask
3	Please clarify whether the Kubernetes Worker nodes classified under GCC will host only GCC/platform services or also Consumption Layer workloads such as NiFi, Kafka, Airflow, Spark, Trino, etc.	This directly affects CPU/RAM sizing and segregation.
4	If Consumption Layer workloads are deployed on Kubernetes, please identify the dedicated worker-node capacity allocated for Consumption Layer workloads.	The current sheet appears to show K8 workers under GCC, so consumption compute capacity needs to be clearly accounted for.
5	Please clarify whether the GCC and Consumption Layer are physically isolated, logically isolated through Kubernetes namespaces/node pools, or only resource-separated at VM level.	Important for security, performance and resource isolation.
6	Please confirm whether GCC components are separately sized for Production-DC, Production-DR, UAT and Pre-Prod, or whether any GCC services are shared across environments.	The sheet shows GCC components across multiple environments; we need to understand why.
7	MinIO/Object Storage is classified under Consumption Layer. Please confirm whether MinIO is intended to serve as the Data Lake/Object Storage layer or as shared infrastructure storage for the GCC platform.	Classification depends on its intended role.
8	Please provide the CPU, RAM, storage and GPU allocation attributable specifically to GCC and Consumption Layer for each environment.	Helps independently validate the sizing.
9	Please provide the expected Kubernetes node-pool/cluster architecture, including GCC node pools and Consumption node pools, if applicable.	To validate how workloads will actually be deployed.
10	Please clarify whether the Consumption Layer services will run on dedicated compute resources or share the GCC Kubernetes worker resources.	Shared resources can create contention and affect sizing.
11	Please confirm the HA/DR requirements for each GCC and Consumption Layer component and indicate which components require active-active, active-passive or single-instance deployment.	Directly impacts quantity and sizing.

#	Question to Vendor	Why we should ask
12	Please explain the basis for the quantity of K8 Master and K8 Worker nodes proposed for each environment.	We need to verify whether the node count follows HA and workload requirements.
13	Please provide the resource overhead considered for Kubernetes, monitoring, logging, service mesh and other platform services in the sizing.	Prevents under-sizing of actual usable compute capacity.
14	Please confirm whether Monitoring is only monitoring GCC infrastructure or also all Consumption Layer workloads and applications.	Monitoring scope affects CPU/RAM/storage requirements.
15	Please confirm whether the same Monitoring and Keycloak instances are intended to serve multiple environments or whether separate instances are required per environment.	Avoids unnecessary duplication or, conversely, inadequate isolation.

● The 5 questions I would definitely send

If you don't want to overwhelm the vendor, send these:

1. Please provide the detailed architectural mapping and justification for the segregation of all components between GCC and Consumption Layer.
2. Keycloak is currently classified under Consumption Layer. Please clarify the rationale for this classification and confirm whether Keycloak is intended to function as a shared IAM/platform service across the solution.
3. Please confirm whether the Kubernetes Worker nodes classified under GCC will host only GCC/platform services or also Consumption Layer workloads. If Consumption workloads are hosted on Kubernetes, please provide the dedicated worker-node capacity allocated for them.
4. Please clarify whether GCC and Consumption Layer are physically isolated or logically isolated through dedicated Kubernetes clusters/node pools, namespaces or VM/resource allocation.
5. Please clarify the rationale for sizing separate GCC components across PROD-DC, PROD-DR, UAT and Pre-Prod, and confirm which GCC services are environment-specific versus shared.

My biggest concern is Question #3. If the vendor says “all Consumption services will run on the same K8 Worker nodes already classified under GCC”, then the current GCC/Consumption segregation in the sizing sheet is **not a clean resource segregation**. The vendor should then show exactly how much CPU/RAM of those workers is allocated to GCC versus Consumption workloads.



check if all sensitive data carrying components are in GCC?

No. Based on the sizing sheet you shared, the sensitive-data-carrying components are **NOT** all placed under GCC. In fact, most of the major data-bearing components are currently classified under the Consumption Layer.

1. What I see in your sheet

Component	Current classification	Carries/processes sensitive data?	My observation
K8 Master	GCC	✗	Platform/control-plane component
K8 Worker	GCC	⚠ Depends on workloads	Could host sensitive workloads
Monitoring	GCC	⚠ Potentially	Logs/metrics can contain sensitive information
Keycloak	Consumption	⚠ Yes – identity/auth data	Should be reviewed
Apache NiFi	Consumption	✓ Yes	Ingests/moves enterprise data
Kafka	Consumption	✓ Yes	Transports/retains data in topics

Component	Current classification	Carries/processes sensitive data?	My observation
Airflow	Consumption	⚠️ Potentially	Workflow metadata/connections/secrets
Spark Master/Worker	Consumption	✅ Yes	Processes large datasets
Trino Coordinator	Consumption	✅ Yes	Queries data sources
OpenSearch	Consumption	✅ Yes	Stores/searches indexed data
PostgreSQL	Consumption	✅ Yes	Database
MongoDB	Consumption	✅ Yes	Database
MinIO / Object Storage	Consumption	✅ Yes – major concern	Likely Data Lake/object data
Milvus	Consumption	✅ Yes	Stores vector/embedding data
Neo4j	Consumption	✅ Yes	Stores graph/business data
GPU/H200	Consumption	⚠️ Processes sensitive data	AI/ML workloads may process sensitive datasets

2. The important point

If by **GCC** you mean the **secure/core infrastructure zone where sensitive data is supposed to reside**, then the current segregation is **not aligned with that principle**.

The sheet puts these major data-bearing components in **Consumption Layer**:

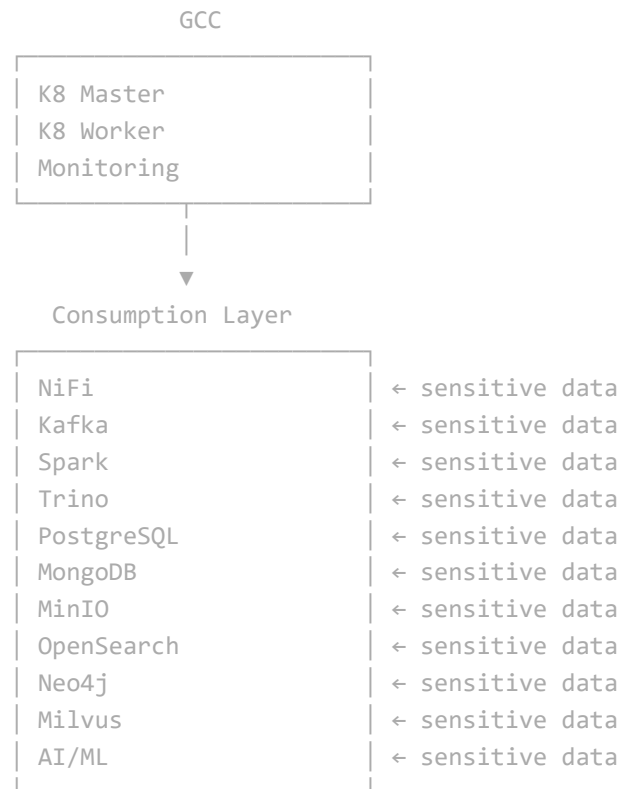
MinIO + PostgreSQL + MongoDB + OpenSearch + Neo4j + Milvus + Kafka

And the processing components:

NiFi + Spark + Trino + Airflow

are also in Consumption.

So the current architecture appears to be:



Therefore, you should not simply accept the statement that "GCC contains the sensitive data components." The sizing sheet does not show that.

3. But there is an important architectural distinction

I would not automatically say that every data component must be moved to GCC.

It depends on what the RFP means by GCC.

If GCC is intended to be the **General Compute Cluster / shared infrastructure layer**, then it is actually reasonable for:

- MinIO

- PostgreSQL
- MongoDB
- Kafka
- Spark
- Trino
- OpenSearch
- etc.

to be part of the **Consumption/Data platform** layer.

The key question is instead:

Where is the security boundary for sensitive data?

You need to establish whether the architecture provides **secure isolation and protection of sensitive data**, irrespective of whether the component is labelled GCC or Consumption.

4. I would specifically question the vendor on this

Ask:

Please identify all components in the proposed architecture that will store, process, transmit or temporarily cache sensitive/PII/financial/business-critical data, and provide their corresponding security zone (GCC/Consumption), encryption mechanism, access-control mechanism and data-at-rest/data-in-transit protection.

And specifically:

Please confirm whether sensitive data will reside within the Consumption Layer components such as MinIO, PostgreSQL, MongoDB, Kafka, OpenSearch, Neo4j and Milvus. If yes, please explain the security isolation between GCC and Consumption Layer and how sensitive data is protected.

● One more question I strongly recommend

Please provide a data-flow diagram showing the movement of sensitive data from ingestion (NiFi/Kafka) → processing (Spark/Trino) → storage (MinIO/Databases) → consumption/analytics, clearly identifying the security zone at each stage.

That question is more useful than simply asking "*Are sensitive components in GCC?*"

Because looking at this sizing sheet, **the sensitive-data journey clearly crosses the Consumption Layer**. The vendor should demonstrate **how that data is protected throughout the journey**, rather than merely classifying infrastructure as GCC or Consumption.



list all questions?

Yes. Based on the **GCC/Consumption segregation, sensitive-data handling, Kubernetes sizing, and the components shown in your sizing sheet**, I would consolidate the vendor queries as follows.

A. GCC vs Consumption Layer Segregation

1. Please provide the detailed mapping of every component/service to GCC or Consumption Layer, along with the architectural justification for the segregation.
2. Please clarify the exact definition and scope of GCC and Consumption Layer in the proposed architecture.
3. Please confirm whether GCC and Consumption Layer are physically isolated, logically isolated, or separated only through VM/resource allocation.
4. Please provide the architecture diagram clearly showing the boundary between GCC and Consumption Layer.
5. Please confirm whether GCC services are dedicated to each environment or shared across PROD-DC, PROD-DR, UAT and Pre-Prod.
6. Please explain the rationale for provisioning separate GCC infrastructure for each environment as shown in the sizing sheet.

B. Sensitive Data – MOST IMPORTANT

7. Please identify all components that will store, process, transmit or temporarily cache sensitive/PII/financial/business-critical data.

8. Please confirm whether sensitive data will reside in Consumption Layer components such as MinIO, PostgreSQL, MongoDB, Kafka, OpenSearch, Neo4j and Milvus.
9. If sensitive data resides in the Consumption Layer, please explain the security boundary and isolation mechanism between GCC and Consumption Layer.
10. Please provide a data-flow diagram showing the movement of sensitive data from ingestion → processing → storage → analytics/consumption, with the security zone identified at each stage.
11. Please confirm whether sensitive data is encrypted at rest and in transit for all applicable GCC and Consumption Layer components.
12. Please specify the encryption standards and key-management mechanism proposed for sensitive data.
13. Please explain how access to sensitive data is controlled across the Consumption Layer.
14. Please confirm whether RBAC, MFA and centralized IAM are implemented for access to sensitive-data components.
15. Please clarify where audit logs relating to sensitive-data access are stored and how long they are retained.

C. Keycloak / IAM

16. Keycloak is currently classified under the Consumption Layer. Please clarify the rationale for this classification.
17. Please confirm whether Keycloak is intended to be a shared IAM/platform service for GCC and Consumption Layer components.
18. If Keycloak is a shared platform/security service, please explain why it has been placed under Consumption Layer instead of GCC.
19. Please confirm the HA/DR architecture and sizing proposed for Keycloak.
20. Please clarify how Keycloak credentials, tokens, secrets and identity information are protected.

D. Kubernetes

21. Please confirm whether the Kubernetes Worker nodes classified under GCC will host only GCC services or also Consumption Layer workloads.

22. If Consumption Layer workloads run on these workers, please provide the CPU/RAM capacity specifically allocated to GCC and Consumption workloads.
23. Please provide the proposed Kubernetes cluster/node-pool architecture.
24. Please clarify whether dedicated node pools are proposed for GCC and Consumption workloads.
25. Please explain how resource isolation is enforced between GCC and Consumption workloads.
26. Please confirm whether sensitive workloads can be restricted to dedicated Kubernetes nodes using node affinity, taints/tolerations or equivalent controls.
27. Please provide the basis for the number of Kubernetes Master and Worker nodes proposed for each environment.
28. Please clarify the CPU/RAM overhead considered for Kubernetes control-plane and platform services in the sizing.

E. Data Storage Components

29. MinIO/Object Storage is classified under Consumption Layer. Please confirm whether MinIO is the primary Data Lake/Object Storage platform.
30. Please clarify whether MinIO will store sensitive/PII/business-critical data.
31. Please explain the HA, replication and DR architecture proposed for MinIO.
32. Please clarify how data isolation is maintained between different environments and workloads within MinIO.
33. Please provide the retention, backup and recovery strategy for MinIO and other databases.
34. Please confirm whether PostgreSQL, MongoDB, OpenSearch, Neo4j and Milvus will contain production sensitive data.
35. Please provide the HA/DR and backup strategy for each of these data stores.

F. Data Processing & Movement

36. Please confirm whether Apache NiFi will ingest sensitive data from source systems.
37. Please explain how sensitive data is secured while being transferred through NiFi and Kafka.
38. Please clarify Kafka topic-level security, authentication, authorization and encryption mechanisms.
39. Please confirm whether Kafka retains sensitive data and specify the proposed retention period.
40. Please explain how sensitive data is protected while being processed by Spark.

- 41. Please clarify whether Spark temporary/staging data can contain sensitive information and how it is secured/deleted.
- 42. Please confirm whether Trino can directly query sensitive datasets and explain the access-control mechanism.
- 43. Please explain how Airflow protects database credentials, connection strings, API keys and other secrets.

G. Monitoring & Logging

- 44. Monitoring is classified under GCC. Please confirm whether it will monitor both GCC and Consumption Layer components.
- 45. Please clarify whether monitoring/logging systems may capture sensitive data, PII, credentials, tokens or business information.
- 46. If sensitive information can appear in logs, please explain the masking/redaction mechanism.
- 47. Please provide the log retention and storage sizing assumptions.
- 48. Please clarify who can access monitoring and application logs and how access is controlled.

H. DC / DR / UAT / Pre-Prod

- 49. Please confirm whether the same GCC and Consumption architecture is replicated across PROD-DC and PROD-DR.
- 50. Please provide the DR architecture and identify which components are active-active, active-passive or backup-based.
- 51. Please provide the RPO and RTO considered for each sensitive-data component.
- 52. Please clarify whether UAT and Pre-Prod contain production/sensitive data or masked/anonymized data.
- 53. If production data is used in UAT/Pre-Prod, please explain the masking/anonymization mechanism.
- 54. Please confirm whether sensitive data is permitted to move between PROD, UAT and Pre-Prod environments.

I. Sizing Validation

55. Please provide the CPU, RAM, SSD/HDD and GPU requirement separately for GCC and Consumption Layer for each environment.
56. Please provide the sizing calculation/basis for every component rather than only the final quantity.
57. Please specify the expected workload, transaction volume, data volume and growth assumptions used for sizing.
58. Please clarify the utilization threshold and headroom considered in the CPU and RAM sizing.
59. Please confirm whether HA, failover and N+1 capacity have been included in the sizing.
60. Please clarify whether backup, replication, temporary/staging and recovery storage have been included in the storage sizing.

My TOP 10 questions to send first

If you want to keep the vendor query list short, I would send these 10:

1. Please provide the detailed mapping and architectural justification of every component between GCC and Consumption Layer.
2. Please clearly define the security boundary between GCC and Consumption Layer and confirm whether the segregation is physical, logical or resource-based.
3. Please identify all components that will store, process or transmit sensitive/PII/business-critical data and identify their respective security zones.
4. Please confirm whether sensitive data will reside in Consumption Layer components such as MinIO, PostgreSQL, MongoDB, Kafka, OpenSearch, Neo4j and Milvus, and explain how such data is protected.
5. Keycloak is classified under Consumption Layer. Please clarify the rationale and confirm whether it is intended to be a shared IAM/platform service.
6. Please confirm whether Kubernetes Worker nodes classified under GCC will host Consumption Layer workloads. If yes, provide the resource allocation between GCC and Consumption workloads.
7. Please provide the Kubernetes node-pool/cluster architecture and explain how GCC and Consumption workloads are isolated.
8. Please provide the sensitive-data flow from ingestion → Kafka/NiFi → processing → storage → analytics/consumption, including security controls at each stage.
9. Please confirm encryption, IAM/RBAC, audit logging, backup, retention and DR mechanisms for all sensitive-data components.

10. Please provide the CPU/RAM/storage/GPU sizing calculation separately for GCC and Consumption Layer, including HA, DR, utilization and growth headroom assumptions.

One key point for your review

I would **not** frame the query as “all sensitive data components should be in GCC.” That may not be architecturally correct.

The stronger query is:

“Identify where sensitive data resides and demonstrate how it is securely isolated and protected irrespective of whether the component is classified under GCC or Consumption Layer.”

That will force the vendor to explain the **actual security architecture**, rather than simply defending the labels in the sizing sheet.



ChatGPT can make mistakes. Check important info.